



Comau's Racer3
(Image courtesy: www.luzitellidaniell.com)

ing the excess away before adding the next one.

The team demonstrated that this device could build 14 different classes of small molecules. Essentially, it transforms small molecule synthesis from a highly individualised, customised process, to a more generalised automated approach, which has the potential to speed up new drug development.

Adam and Eve. One step ahead of being helpers in the lab, robot scientists like Adam and Eve are ready to develop drugs themselves. According to a University of Cambridge press report, robot scientists can, "automatically develop and test hypotheses to explain observations, run experiments using laboratory robotics, interpret the results to amend their hypotheses and then repeat the cycle, automating high-throughput hypothesis-led research."

Adam, a robot scientist developed in 2009 by researchers at Universities of Aberystwyth and Cambridge, both in the UK, became the first machine to independently discover new scientific knowledge. In its wake comes Eve, a robot recently developed by the same team, based at University of Manchester, UK.

Eve can automate early-stage

drug design by systematically testing each member from a large set of compounds in the standard brute-force way of conventional mass screening these compounds against assays (tests) designed to be automatically engineered and using statistics and machine learning to predict new structures that might score better against the assays.

Future versions of the robot might also be able to synthesise these recommended compounds. It is hoped that Eve can help identify new drug candidates for malaria and neglected tropical diseases such as African sleeping sickness and Chagas disease, which are killing millions of people every year.

Prof. Ross King, from Manchester Institute of Biotechnology at University of Manchester, says in a media report, "Every industry now benefits from automation, and science is no exception. Bringing in machine learning to make this process intelligent, rather than just a brute-force approach, could greatly speed up scientific progress and potentially reap huge rewards."

Real helping hands

Known more scientifically as collaborative robots, these are a rising new clan of robots that can truly work side-by-side with humans. Imagine a robot lending a real helping hand, lifting a heavy object into a truck, the way you manually guide it to, just as two humans would lift it! The current generation of agile and safe industrial robots are really breaking out of caged production lines and working besides humans.

Experts say that, when man and robot share the same workspace, it creates fruitful synergies and productivity benefits by combining the strengths and benefits of the machine, such as reliability, durability and repetition accuracy, with human strengths such as dexterity and flexibility. However, this class of robots also requires special designs, in terms of greater flexibility, receptiveness to

ad-hoc instructions and enhanced safety measures. For example, it must be possible for the robot to assess its co-worker's position and classify it into safe zones, warning zones and unsafe zones in real-time. The robot should also be able to automatically adjust its own position to always revert to safe zone.

New components such as Safety-EYE, a 3D camera system that helps monitor protected fields and detection zones in several dimensions, aid in implementing human-robot collaboration.

Comau's Racer3 is a recent example of a robot capable of working besides humans. It is a high-strength aluminium 6-axis articulated robot that can be mounted on benches, walls, ceilings or inclined supports. It has been designed for out-of-the-cage use in industries like food and beverage, electronics, plastics and metalworking. Speed and flexibility are what Racer3 really excels in.

According to the company's official press release, "Racer3 is a natural contortionist that can bend like a snake until it assumes the shape of a scissor, surpassing the flange at axis 1. Or, it can close like a book, bringing the wrist toward the body to rotate the axis 1 at maximum speed while avoiding collisions with anything in the surrounding space. With a cycle time of less than 0.36 seconds for a 1kg pick-and-place cycle, the rigid construction and stable base ensures high precision and repeatability."

Fanuc's CR-35iA stands apart from common collaborative robots in the market, which are usually small payload varieties. This particular helper is capable of handling 35kg payloads, which according to the company is an industry-first. A soft green foam cover helps protect humans working around the robot, which reportedly meets ISO 10218-1:2011 robotic safety requirements and is designed to work alongside humans in operations such as machine tending, handling heavy pay-